

Implementation of a Basic Blockchain with PoW and Merkle Patricia Tries

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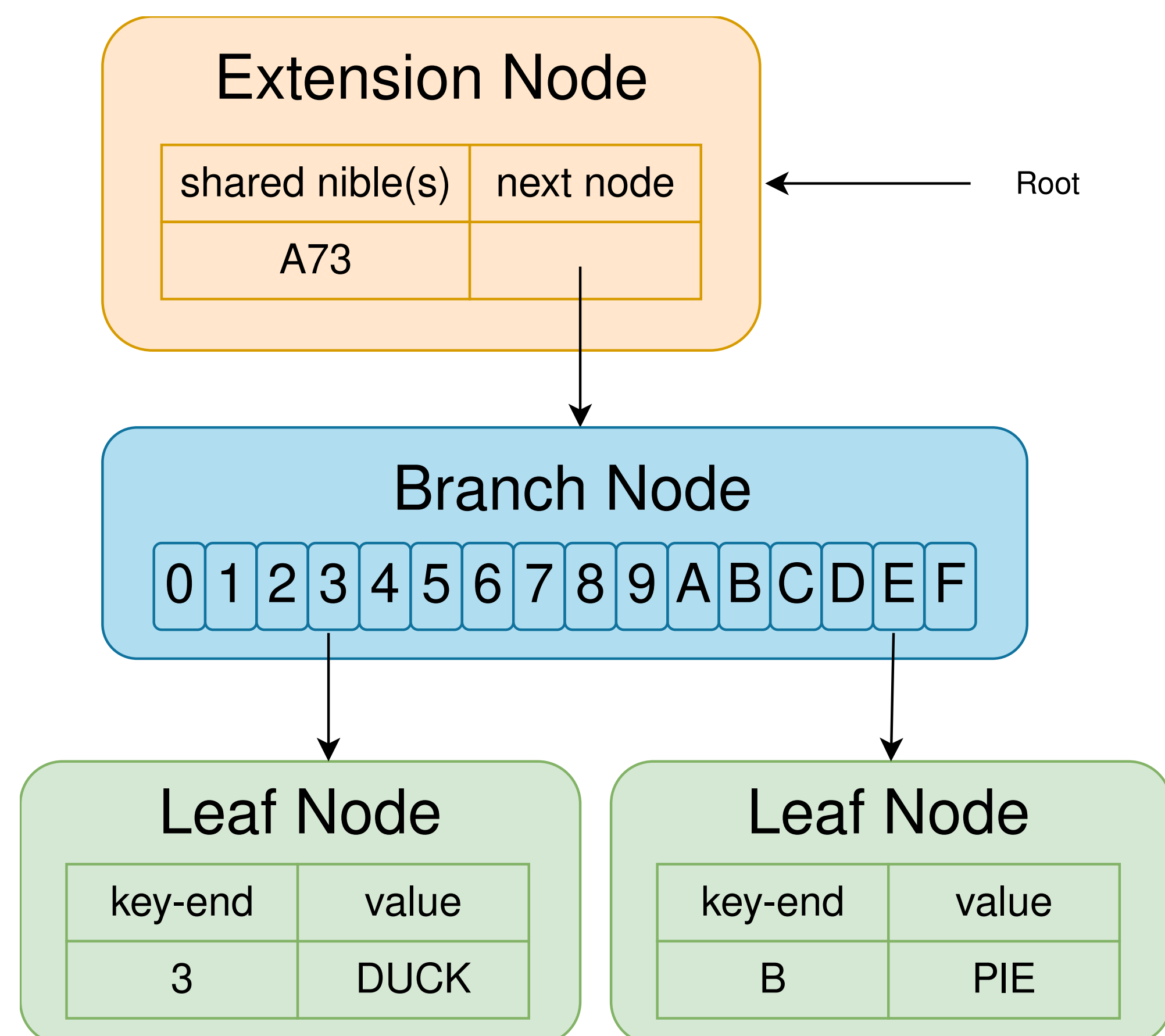
1. Objectives

1. Blockchain from scratch in Go
2. Implementation of Proof-of-Work (PoW) as consensus mechanism
3. Build a database structure based on Merkle Patricia Trie (MPT)

2. Capabilities

- Block immutability
- Lightweight Verification (MPT Proofs)
- Persistent Storage
- Debugging tools and tests

3. Merkle Patricia Trie (MPT) State

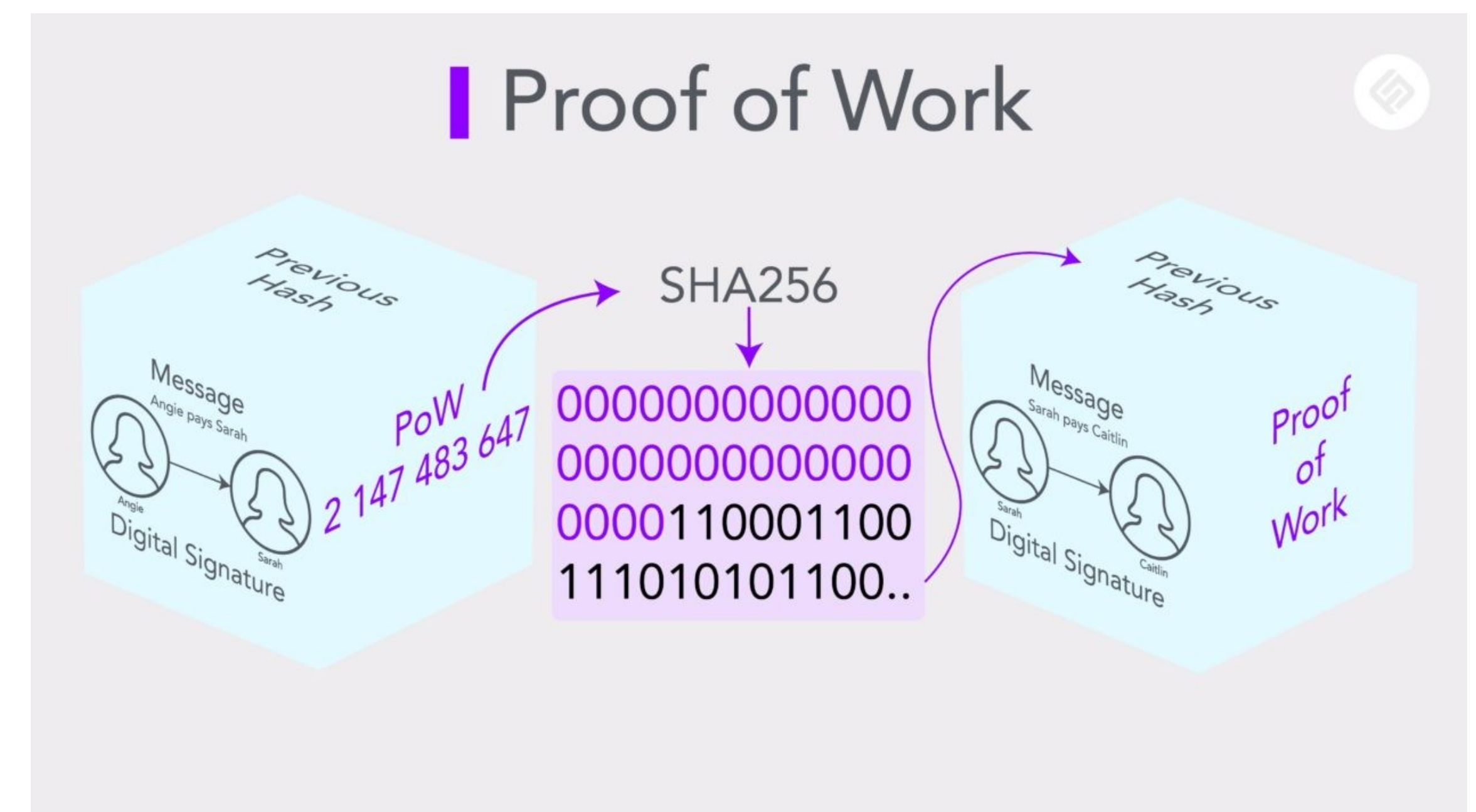


Key	Value
0xA7333	DUCK
0xA73EB	PIE

4. Consensus (PoW)

$$\text{SHA256}(\text{BlockData} \parallel \text{Nonce}) < \text{Target}$$

$$\text{Target} = 1 \ll (\text{HashLength} - \text{Difficulty})$$



5. Class diagram

